DGMS 2019 • SECTION J: ORE GEOLOGY & ECONOMIC GEOLOGY

Question 77: Algoma-Type vs. Superior-Type Banded Iron Formations

Official Examination Guide • Precambrian Secular Evolution & Depositional Tectonic Settings

OFFICIAL EXAMINATION QUESTION

Consider the following statements:

1. Algoma-type iron formations are sedimentary units in sequences of submarine volcanic rocks in greenstone belts, most commonly of Cambrian age.

2. Superior-type iron formations occur in sedimentary basins of marine strata of late Archaean or Palaeoproterozoic age.

Which of the statements given above is/are correct?

(a) 1 only

(b) Both 1 and 2

(c) 2 only CORRECT ANSWER

(d) Neither 1 nor 2

1. Executive Summary & Core Principle

Banded Iron Formations (BIFs) are chemical sedimentary rocks composed of alternating millimeter- to centimeter-scale layers of iron-rich minerals (magnetite, hematite, siderite, greenalite) and silica-rich chert. Following the classic

classification by **G.A. Gross (1965, 1980)**, Precambrian iron formations are subdivided primarily into:

- **Algoma-Type Iron Formations:** Deposited in volcanic-arc and back-arc environments within **Archean greenstone belts (>2.5 Ga, mostly 2.7–3.5 Ga)**. They are intimately interbedded with submarine volcanic rocks (pillowed tholeiitic basalts, andesites, felsic tuffs) and volcanogenic turbiditic greywackes. They are typically of limited aerial extent (a few kilometers long, tens of meters thick) and originated from local submarine volcanic exhalative hydrothermal venting. **They are definitively ARCHEAN, NOT Cambrian!** The Cambrian (541–485 Ma) is a Phanerozoic system during which typical BIFs were completely absent from Earth's oceans.
- **Superior-Type Iron Formations:** Deposited in broad, shallow-marine epicontinental shelf basins and passive continental margins during the **Late Archean to Paleoproterozoic (2.45 to 1.8 Ga)**. They are associated with platformal sedimentary strata (quartzites, dolostones, black shales) without direct volcanic rock associations. They extend over hundreds of square kilometers and represent basin-scale oxidation of upwelling ferruginous deep oceanic waters during the Great Oxidation Event (GOE).

Because Statement 1 incorrectly assigns a *Cambrian* age to Algoma-type iron formations, Statement 1 is false. Statement 2 accurately describes the Late Archean to Paleoproterozoic age and marine sedimentary association of Superior-type BIFs. Thus, the correct option is **(c) 2 only**.

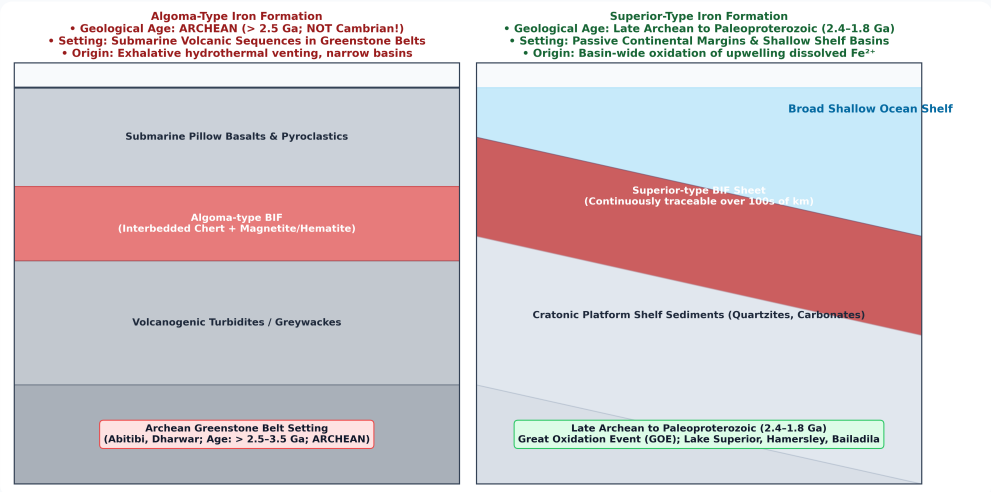


Figure 77.1: Comparative tectonic, stratigraphical, and temporal models for Algoma-type and Superior-type Banded Iron Formations (BIFs), highlighting the Archean volcanic greenstone association of Algoma-type vs. the Paleoproterozoic

*passive shelf setting of Superior-type BIFs (Grounded in Gross 1980 and Misra, *Understanding Mineral Deposits*).*

2. Detailed Comparative Classification of Iron Formations (Gross 1980)

Attribute	Algoma-Type Iron Formation	Superior-Type Iron Formation
Geological Age	**Archean (> 2.5 Ga; typically 2.7–3.5 Ga)**	**Late Archean to Paleoproterozoic (2.45–1.8 Ga)**
Tectonic Setting	Submarine volcanic arcs, back-arc basins, rifted greenstone belts	Stable continental shelves, cratonic margins, epicontinental basins
Associated Lithologies	Submarine pillow basalts, felsic pyroclastics, greywackes, turbidites	Quartz arenites, dolomites, cherts, black carbonaceous shales (no direct volcanics)
Areal Extent & Thickness	Narrow, lenticular, localized (strikes 1–10 km, thickness <100 m)	Massive, laterally continuous (hundreds of km, thickness 100–500 m)
Iron Source & Genesis	Direct submarine volcanic-exhalative hydrothermal fluids	Hydrothermal upwelling into stratified ocean; oxidized by early atmospheric O_2 (GOE)
Global Examples	Algoma District (Ontario), Abitibi belt (Canada), Bababudan/ Kudremukh (Dharwar, India)	Mesabi & Gunflint Ranges (Lake Superior, USA/Canada), Hamersley Basin (Australia), Noamundi/Bailadila (India)

3. Secular Evolution of BIFs and the "Cambrian Fallacy"

The temporal distribution of iron formations in Earth history is one of the most famous milestones in Precambrian geochemistry:

1. **Archean (3.8–2.5 Ga):** Dominated by Algoma-type BIFs. Under an anoxic atmosphere and ocean, dissolved ferrous iron (Fe^{2+}) was abundant in seawater. Local hydrothermal vents provided iron and silica, which precipitated as iron-bearing sediments.
2. **Paleoproterozoic (2.45–1.8 Ga):** Peak of Superior-type BIFs. The advent of oxygenic photosynthesis by cyanobacteria triggered the **Great Oxidation Event (GOE)** (~2.4 Ga). When upwelling deep ferrous-rich marine waters contacted oxygenated surface waters, dissolved Fe^{2+} oxidized to insoluble Fe^{3+} oxyhydroxides, blanketing continental shelves with colossal BIF deposits.
3. **Mesoproterozoic to Neoproterozoic Hiatus (1.8–0.8 Ga):** After ~1.8 Ga, oceans became euxinic (sulfidic, "Canfield Ocean") or deep waters became oxygenated, stripping dissolved Fe^{2+} from seawater and causing the global disappearance of BIFs.
4. **Neoproterozoic Glaciogenic (Rapitan-type, ~0.7 Ga):** Re-appeared briefly during "Snowball Earth" events (Rapitan Group, Canada; Sturtian glaciation), where global sea-ice cover caused ocean anoxia, allowing Fe^{2+} accumulation until ice melted and rapid aeration precipitated iron.
5. **Phanerozoic (Cambrian to Recent, 541 Ma – Present):** Banded Iron Formations **NEVER formed** during the Cambrian! Phanerozoic sedimentary iron ores are exclusively minette-type or Clinton-type *ironstones* (oolitic goethite-chamosite-hematite beds), which are mineralogically, texturally, and genetically distinct from BIFs.

4. Systematic Evaluation of Exam Statements

- **Statement 1 is INCORRECT:** "Algoma-type iron formations are sedimentary units in sequences of submarine volcanic rocks in greenstone belts, most commonly of Cambrian age."
  Why: The first part of the statement is true (they are sedimentary units in submarine volcanic sequences in greenstone belts), but the age clause

—*"most commonly of Cambrian age"*—is factually false. They are Archean (>2.5 Ga).

- **Statement 2 is CORRECT:** "Superior-type iron formations occur in sedimentary basins of marine strata of late Archaean or Palaeoproterozoic age."
  Why: This is completely accurate. Superior-type BIFs represent shelf sedimentary sequences that peaked between 2.45 Ga and 1.8 Ga in the Paleoproterozoic.

> **Key Takeaway:** Algoma-type BIF = Archean submarine volcanics in greenstone belts; Superior-type BIF = Paleoproterozoic marine shelf sediments. Hence, **(c) 2 only** is the correct choice.

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